Arcs complete outside a prescribed conic

A prescribed-hole defect identity and a certified classification over $\mathbb{F}_{16}$

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Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$. We study arcs $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ whose secants cover every point outside $A \cup \mathcal{C}$, and write $\rho_{\mathcal{C}}(q)$ for their minimum size. Starting from the classical first two equations for the secant-index distribution of an arc, we prove an exact defect identity for an arbitrary prescribed set of points that may remain uncovered, together with equality and stability criteria. Its conic specialization gives

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{\lfloor k/2 \rfloor}$$

for every $\mathcal{C}$-complete k-arc, together with an exact equality criterion and a quantitative stability estimate. In particular, for every prime power q,

$$\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.$$

Conversely, if H is any prescribed set and an ordinary complete b-arc satisfies $b|H| < q^2 + q + 1$, projective averaging moves it off H; the conic case gives $\rho_{\mathcal{C}}(q) \leq t_2(2,q)$ whenever $t_2(2,q) \leq q$. We also give tangent and parity constraints in even characteristic. A sharp finite-field evaluation dichotomy packages the quadratic obstruction for arbitrary feature maps, and the arc–MDS dictionary turns the defect identity into an exact syndrome-leader collision identity. Directly checkable witnesses give

$$\rho_{\mathcal{C}}(8) = \rho_{\mathcal{C}}(9) = \rho_{\mathcal{C}}(11) = 6, \qquad \rho_{\mathcal{C}}(16) = 9.$$

A classification of all 2633 projective eight-arcs over $\mathbb{F}_{16}$ proves the stronger statement that no nonzero quadratic can contain an eight-arc's ordinary-uncovered locus while avoiding the arc. A supplementary verifier checks the witnesses, while a companion Lean formalization proves the theorem chain and kernel-checks the local classification, quadratic obstruction, projective reduction, and exact $q = 16$ conclusion. For the six-point $q = 11$ witness it certifies the no-conic premise which, by the classical normal-rational-curve dictionary, makes the associated $[6,3,4]_{11}$ MDS code projectively non-GRS; it also proves covering radius three, the exact coset and extension spectra, and the icosahedral simultaneous-extension complex.

1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. An arc is complete when it cannot be enlarged while remaining an arc; equivalently, every point outside it lies on a secant determined by two of its points. Complete arcs and the closely related problem of finding small saturating sets have a substantial literature; see, for example, Ball [3], Kim–Vu [12], and Nagy [15].

Fix a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2,q)$. We impose the arc condition but allow the points of $\mathcal{C}$ to remain uncovered.

Definition 1.1. An arc $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ is *$\mathcal{C}$-complete* if every point of

$$\mathrm{PG}(2,q) \setminus (A \cup \mathcal{C})$$

lies on a secant of A. Define

$$\rho_{\mathcal{C}}(q) = \min\{|A| : A \text{ is a } \mathcal{C}\text{-complete arc}\}.$$

For an arbitrary arc disjoint from $\mathcal{C}$, its uncovered locus is

$$\mathcal{U}_{\mathcal{C}}(A) = \mathrm{PG}(2,q) \setminus \left(A \cup \mathcal{C} \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right).$$

Thus A is $\mathcal{C}$-complete exactly when $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$.

The minimum in Definition 1.1 is attained. Indeed, in the finite family of arcs disjoint from $\mathcal{C}$, any member of maximum cardinality is maximal under insertion and hence is $\mathcal{C}$-complete. The same argument works for every prescribed hole set.

Equivalently, A is maximal among arcs in the point set $\mathrm{PG}(2,q) \setminus \mathcal{C}$. All nonsingular conics in $\mathrm{PG}(2,q)$ are projectively equivalent, so the numerical value of $\rho_{\mathcal{C}}(q)$ does not depend on the chosen conic.

The problem differs from two nearby standard notions. A saturating set need not be an arc and has no prescribed exceptional locus. An almost-complete subset of a conic, in the sense of Bartoli–Davydov–Marcugini–Pambianco [5], consists of selected points *on* the conic whose chords cover the complementary points. Ng–Wild's arcs covering a line [16] prescribe a one-dimensional set that must be covered, rather than an exceptional set that may remain uncovered. Here the selected arc is required to avoid the conic.

There is also classical work on *complete exterior sets* of a conic: sets of $(q+1)/2$ exterior points whose pairwise joins are exterior lines; see Blokhuis–Seress–Wilbrink [6, pp. 143, 146], who report that, up to isomorphism, one of the two $q = 11$ configurations is a six-arc. This is the configuration described thirty-five years earlier by Edge [7, §§29–32], who named it a *Clebsch hexagon* after its real-plane antecedent. This is a different condition from prescribed-hole completeness: it constrains how the selected points meet the conic, whereas Definition 1.1 requires their secants to cover the complementary off-conic locus. The $q = 11$ witness below has all of its secants exterior to the conic, making this literature directly relevant even though the paper's completeness condition is different. In uncovered-locus notation, the classical exterior-set condition gives only $\mathcal{C}(\mathbb{F}_{11}) \subseteq U(A)$. The reverse inclusion used below follows by combining Dye's ten-Brianchon-point result with the elementary chord-defect count; it is not stated in the BSW paper.

The point index with respect to an arc, and the first two double-counting equations for its distribution, are classical; compare Hirschfeld [11, Chapter 9] and Ball [3]. The use of secant counts to bound the smallest complete (k,n)-arcs is also classical; for a recent explicit lower bound see Alabdullah–Hirschfeld [1]. The main observation of this note is that, after splitting those equations over a prescribed exceptional set and its complement, the inequalities normally used to control overlap have an exact remainder. This produces:

(i) an exact prescribed-hole defect identity and equality criterion;

(ii) a quantitative stability bound for near equality;

(iii) a strengthened lower bound with additive asymptotic term $3/2$;

(iv) a projective-averaging upper-bound transfer from ordinary complete arcs;

(v) characteristic-two nucleus constraints;

(vi) an exact evaluation-avoidance dichotomy and a coding-theoretic form of the prescribed-hole defect; and

(vii) small finite-field results whose upper bounds are independently verifiable from explicit coordinates.

Thus the classical scaffolding is the secant-index equations, the Lunelli–Sce $\sqrt{2q}$ scale, the arc–MDS dictionary, and standard saturating-set terminology. The new claims isolated here are the prescribed-hole parameter, the exact remainder after the exceptional-set split (including its equality and stability consequences), and the use of the ordinary-uncovered locus as an evaluation obstruction. The defect identity is deliberately an exact accounting theorem derived from the two classical moments; its content is the identified nonnegative remainder and the consequences obtained by retaining it, not a claim that the underlying moments are new.

A targeted literature search across complete and saturating arcs, almost-complete conic subsets, prescribed-line coverage, arcs and quadrics, and the published classification of arcs in $\mathrm{PG}(2,16)$ did not locate Definition 1.1 or the exact defect identity below in this form. This observation is not a priority certificate: the proofs are elementary consequences of classical index equations, and terminology involving relative saturation or holes may occur elsewhere.

2 The classical secant equations

Throughout the remainder of the paper, $q \geq 3$ and $k \geq 3$. Let Π be a projective plane of order q, let A be a k-arc in Π, and let $r_A(x)$ be the number of secants of A through $x \in \Pi \setminus A$. We usually write $r(x)$. Put

$$N = \binom{k}{2}, \qquad m = \left\lfloor \frac{k}{2} \right\rfloor .$$

Lemma 2.1. *For every $x \notin A$, one has $r(x) \leq m$.*

Proof. Distinct secants through x cannot share an endpoint in A. Indeed, two lines through the same pair x, a coincide. Hence the secants through x use pairwise disjoint pairs of points of A, of which there are at most $\lfloor k/2 \rfloor$. □

Proposition 2.2 (Classical first and second index equations)**.** *For every k-arc A in a projective plane of order q,*

$$\sum_{x \notin A} r(x) = \binom{k}{2}(q-1), \tag{1}$$

$$\sum_{x \notin A} \binom{r(x)}{2} = 3\binom{k}{4}. \tag{2}$$

Proof. There are $\binom{k}{2}$ secants. Each contains $q+1$ points, exactly two of them in A, and therefore contributes $q-1$ incidences with points outside A. This proves (1).

For (2), count unordered pairs of distinct secants through a common external point. By Lemma 2.1, their four endpoints in A are distinct. Conversely, a four-subset of A has three partitions into two unordered pairs. Each partition gives two secants, which meet in a unique point because the plane is projective. Their intersection is outside A: otherwise one of the two secants would contain three points of the arc. Thus every four-subset contributes exactly three. $\square$

Only the projective-plane axioms and the common line size $q+1$ enter these proofs. In particular, the second equation need not hold in a general finite linear space where two lines may be disjoint.

3 An exact defect identity for prescribed holes

The next theorem is stated for an arbitrary exceptional point set. The conic enters only in later specializations.

Let $\mathcal{H} \subseteq \Pi \setminus A$ be any set of points allowed to remain uncovered, and define

$$\begin{aligned} V_{\mathcal{H}}(A) &= \Pi \setminus (A \cup \mathcal{H}), \\ \mathcal{X}_{\mathcal{H}}(A) &= \{x \in V_{\mathcal{H}}(A) : r(x) > 0\}, \\ \mathcal{U}_{\mathcal{H}}(A) &= V_{\mathcal{H}}(A) \setminus \mathcal{X}_{\mathcal{H}}(A), \\ I_{\mathcal{H}}(A) &= \sum_{y \in \mathcal{H}} r(y). \end{aligned}$$

Thus $\mathcal{X}_{\mathcal{H}}(A)$ is the covered required locus, and $\mathcal{U}_{\mathcal{H}}(A)$ is the uncovered required locus.

Splitting Proposition 2.2 over $V_{\mathcal{H}}(A)$ and $\mathcal{H}$ gives

$$\sum_{x \in V_{\mathcal{H}}(A)} r(x) = N(q-1) - I_{\mathcal{H}}(A), \tag{3}$$

$$\sum_{x \in V_{\mathcal{H}}(A)} \binom{r(x)}{2} + \sum_{y \in \mathcal{H}} \binom{r(y)}{2} = 3\binom{k}{4}. \tag{4}$$

Theorem 3.1 (Prescribed-hole defect identity)**.** *Let A be a k-arc in a projective plane of order q, let $\mathcal{H} \subseteq \Pi \setminus A$, and put $m = \lfloor k/2 \rfloor$. Define*

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

Then

$$\boxed{m\Delta_{\mathcal{H}}(A) = \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y \in \mathcal{H}} r(y)(m-r(y)).} \tag{5}$$

In particular, $\Delta_{\mathcal{H}}(A) \geq 0$.

Proof. Only points of $\mathcal{X}_{\mathcal{H}}(A)$ contribute to the sum in (3). Multiplying the definition of $\Delta_{\mathcal{H}}(A)$ by m and using (3) gives

$$m\Delta_{\mathcal{H}}(A) = m \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} r(x) + (m-1)I_{\mathcal{H}}(A) - m|\mathcal{X}_{\mathcal{H}}(A)| - 6\binom{k}{4}.$$

By (4),

$$6\binom{k}{4} = 2 \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} \binom{r(x)}{2} + 2\sum_{y \in \mathcal{H}} \binom{r(y)}{2}.$$

Substitution and termwise simplification yield

$$m(r-1) - 2\binom{r}{2} = (r-1)(m-r),$$
$$(m-1)r - 2\binom{r}{2} = r(m-r),$$

which proves (5). Every term is nonnegative by Lemma 2.1. □

Corollary 3.2 (Coverage and uncovered-locus bounds)**.** *Under the hypotheses of Theorem 3.1,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \le N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}, \tag{6}$$

and

$$|\mathcal{U}_{\mathcal{H}}(A)| \ge |V_{\mathcal{H}}(A)| - N(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{H}}(A)}{m}. \tag{7}$$

Equality holds in either inequality if and only if

$$r(x) \in \{1, m\} \quad (x \in \mathcal{X}_{\mathcal{H}}(A)), \qquad r(y) \in \{0, m\} \quad (y \in \mathcal{H}).$$

Proof. The two inequalities are respectively $\Delta_{\mathcal{H}}(A) \ge 0$ and its rearrangement using $|\mathcal{U}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| - |\mathcal{X}_{\mathcal{H}}(A)|$. The equality statement follows from the vanishing of every nonnegative term in (5). □

For later use, we record the complete-outside consequence before imposing any geometry on the holes.

Corollary 3.3 (Arbitrary prescribed holes)**.** *Let $|\mathcal{H}| = h$. If A is a k-arc disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 + q + 1 - k - h \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}. \tag{8}$$

Proof. Completeness gives $|\mathcal{X}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| = q^2 + q + 1 - k - h$. Substitute this equality into (6). □

The exact remainder also quantifies near equality.

Corollary 3.4 (Quantitative stability)**.** *Assume $m \ge 3$ and put*

$$M = \{x \in \mathcal{X}_{\mathcal{H}}(A) : 2 \le r(x) \le m-1\},$$
$$J = \{y \in \mathcal{H} : 1 \le r(y) \le m-1\}.$$

Then

$$(m-2)|M| + (m-1)|J| \le m\Delta_{\mathcal{H}}(A). \tag{9}$$

Proof. For $2 \le r \le m-1$, $(r-1)(m-r) \ge m-2$. For $1 \le r \le m-1$, $r(m-r) \ge m-1$. Apply these estimates to (5). □

Thus small defect forces almost every multiply covered required point to have the maximum possible multiplicity m, while almost every hole point met by a secant also has multiplicity m. This is the precise stability content behind the equality heuristic.

4 Coding interpretation

We record the standard arc–code dictionary because it makes the defect identity more informative. Choose nonzero column representatives of a k-arc $A \subset \mathrm{PG}(2,q)$ and form a $3 \times k$ matrix H_A. For $k \geq 4$, the code

$$C_A = \{c \in \mathbb{F}_q^k : H_A c^\mathsf{T} = 0\}$$

is a $[k, k-3, 4]_q$ MDS code: every three columns are independent, the columns span $\mathbb{F}_q^3$, and any four columns support a dependence. Conversely, a projective parity-check system for a code with these parameters is a k-arc. This correspondence and its coset-weight consequences are classical; see Davydov–Marcugini–Pambianco [8]. We call such a code *projectively GRS* when, after permuting and scaling columns and applying a projective coordinate change, its projective parity-check columns lie on a normal rational curve; in $\mathrm{PG}(2,q)$ this is a nonsingular conic. This is the standard normal-rational-curve/GRS correspondence [18].

Proposition 4.1 (Syndrome form of relative completeness)**.** *Let $s \neq 0$ have projective direction $x = [s] \in \mathrm{PG}(2,q)$. Then:*

(i) the minimum weight of a word c with $H_A c^\mathsf{T} = s$ is one, two, or three according as $x \in A$, $x \notin A$ lies on a secant of A, or x is ordinarily uncovered;

(ii) if the minimum weight is two, the number of weight-two leaders is exactly $r_A(x)$;

(iii) if the minimum weight is three, the number of weight-three leaders is exactly $\binom{k}{3}$; and

(iv) adjoining x as one more projective parity-check column preserves the MDS property exactly when x is ordinarily uncovered.

Consequently, A is complete outside a prescribed set $\mathcal{H}$ exactly when all its projective distance-three syndrome directions lie in $\mathcal{H}$ (and $A \cap \mathcal{H} = \varnothing$). These directions are deep-hole directions only when covering radius three has separately been established.

Proof. A weight-one representation is a scalar multiple of a selected column. A weight-two representation uses a unique unordered column pair whose projective span is a secant through x; independence of the pair makes its two coefficients unique. If neither case occurs, any three columns form a basis and represent s, so the distance is three. For each three-subset the representation is unique, and none of its coefficients can vanish; hence there are exactly $\binom{k}{3}$ leaders. The extension statement is the same three-column independence test. The leader count also follows as the $d = 4$ case of the general farthest-coset formula in Davydov–Marcugini–Pambianco [8, Theorem 7.7]; the deep-hole/MDS-extension formulation is standard in the Reed–Solomon setting as well [13]. □

Thus $r_A(x)$ is not merely an incidence index: it is the exact number of minimum weight-two leaders of every affine syndrome on direction x. All $q-1$ nonzero scalar multiples of a projective syndrome direction have the same distance and leader count: multiplication by the scalar bijects their coefficient words without changing Hamming support. This is the normalization behind every projective-to-affine factor of $q-1$ below. The first moment counts leaders, the second counts collisions between distinct leader supports, and Theorem 3.1 is an exact leader-collision remainder identity after prescribed directions are removed. In particular, Corollary 3.3 is also a length obstruction for projective codimension-three MDS codes whose distance-three syndrome directions are confined to the prescribed projective set. This is a reformulation of the geometric theorem, not a claim that the arc–MDS or deep-hole dictionaries are new.

5 Specialization to a conic

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$, and let $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ be a k-arc. Since $|\mathcal{C}| = q+1$, there are $q^2 - k$ points outside $A \cup \mathcal{C}$. Write

$$I_{\mathcal{C}}(A) = \sum_{y \in \mathcal{C}} r(y) = \sum_{\ell \text{ an } A\text{-secant}} |\ell \cap \mathcal{C}|.$$

Corollary 5.1 (Conic-loss and uncovered-locus inequalities)**.** *For every such arc,*

$$|\mathcal{U}_{\mathcal{C}}(A)| \geq q^2 - k - \binom{k}{2}(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{C}}(A)}{m}. \tag{10}$$

If A is $\mathcal{C}$-complete, then

$$\boxed{q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{m}.} \tag{11}$$

The first correction in (11) accounts for unavoidable concurrence among secants; the last accounts for secant incidence spent on the exempt conic.

Define

$$L_1(q) = \min\left\{k \geq 3 : q^2 - k \leq \binom{k}{2}(q-1)\right\},$$

$$L_2(q) = \min\left\{k \geq 3 : q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4}\right\}.$$

Then

$$\rho_{\mathcal{C}}(q) \geq L_2(q) \geq L_1(q). \tag{12}$$

The definition of $L_2(q)$ involves only integer arithmetic after multiplication by $\lfloor k/2 \rfloor$.

For calculation, the corrected capacity has the parity-dependent forms

$$\binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} = \begin{cases} \dfrac{k-1}{2}\big(k(q-1) - (k-2)(k-3)\big), & k \text{ even}, \\ \dfrac{k}{2}\big((k-1)(q-1) - (k-2)(k-3)\big), & k \text{ odd}. \end{cases} \tag{13}$$

For comparison, some values are

q	5	8	9	11	13	16	17	19
$L_1(q)$	4	5	5	6	6	7	7	7
$L_2(q)$	4	6	6	6	7	8	8	8

Thus the second-moment correction alone proves, for example, that a $\mathcal{C}$-complete arc in $\mathrm{PG}(2,16)$ has at least eight points. At $q = 5$, the matching lower bound $L_2(5) = 4$ has a four-frame witness: the companion Clebsch analysis checks that its ordinary-uncovered locus is the full rational point set of a nonsingular conic. A queued follow-up will import that witness into the present formal semantics and promote the derived target $\rho_{\mathcal{C}}(5) = 4$ into the verified small-value theorem; the companion owns the broader classification for arcs of sizes four through seven.

Theorem 5.2 (Explicit additive lower bound)**.** *For every prime power q,*

$$\boxed{\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.} \tag{14}$$

Consequently,

$$\liminf_{q\to\infty} (\rho_{\mathcal{C}}(q) - \sqrt{2q}) \geq \frac{3}{2}. \tag{15}$$

Proof. Let k be the size of a $\mathcal{C}$-complete arc and put $s = \sqrt{2q}$. The elementary first-moment inequality $q^2 - k \leq \binom{k}{2}(q-1)$ implies $k \geq s$ for $q \geq 2$: otherwise $k < s \leq q$, while $k^2 < 2q$ gives $\binom{k}{2} < q$, so its right side is smaller than $q(q-1)$ whereas $q^2 - k \geq q(q-1)$. If $k \geq s+2$, then (14) is immediate. Otherwise write $k = s + a$, so $0 \leq a < 2$ and $s \geq 2$.

Since $m = \lfloor k/2 \rfloor \leq k/2$,

$$\frac{6}{m}\binom{k}{4} \geq \frac{12}{k}\binom{k}{4} = \frac{(k-1)(k-2)(k-3)}{2}.$$

Dropping the nonnegative conic term in (11) therefore gives the necessary inequality

$$q^2 - k \leq \frac{k-1}{2}\left(k(q-1) - (k-2)(k-3)\right).$$

After substituting $q = s^2/2$ and $k = s + a$, the right side minus the left side is exactly

$$\frac{2a-3}{4}s^3 + \frac{a^2 - 7a + 10}{4}s^2 + \frac{-3a^2 + 10a - 8}{2}s + \frac{-a^3 + 5a^2 - 8a + 6}{2}.$$

For $0 \leq a \leq 2$, the last three coefficients are at most $5/2$, 1, and 3, respectively. Since $s \geq 2$, their total contribution is at most $4s^2$. The displayed difference is nonnegative, and hence

$$0 \leq \frac{2a-3}{4}s^3 + 4s^2.$$

Dividing by s^2 gives $a \geq 3/2 - 8/s$, which is (14). Letting q tend to infinity gives (15). □

Remark 5.3 (The scale of the conic term)*.* Every secant meets $\mathcal{C}$ in at most two points, so

$$0 \leq I_{\mathcal{C}}(A) \leq 2\binom{k}{2}.$$

At the relevant scale $k \asymp \sqrt{q}$, the term $I_{\mathcal{C}}(A)/m$ is therefore $O(\sqrt{q})$, while the universal overlap correction $6\binom{k}{4}/m$ is $\Theta(q^{3/2})$. Consequently, improving the lower bound on $I_{\mathcal{C}}(A)$ within (11) can be useful for finely balanced finite cases, but cannot improve the additive constant $3/2$ in Theorem 5.2. Any stronger asymptotic result needs an additional mechanism, not merely a spectral estimate for $I_{\mathcal{C}}(A)$.

6 An upper-bound transfer from complete arcs

Let $t_2(2, q)$ denote the smallest size of an ordinary complete arc in $\mathrm{PG}(2, q)$.

Theorem 6.1 (Projective averaging)**.** *Let $H \subseteq \mathrm{PG}(2, q)$ and let B be an ordinary complete b-arc. If*

$$b|H| < q^2 + q + 1, \tag{16}$$

then some projective image of B is disjoint from H and complete outside H.

Proof. Choose a uniformly random projectivity $g \in \mathrm{PGL}(3,q)$. Point transitivity gives

$$\mathbb{E}|gB \cap H| = \frac{b|H|}{q^2+q+1} < 1.$$

Since the intersection size is a nonnegative integer, it is zero for some g. Projectivities preserve ordinary completeness, so gB is complete outside H. □

Corollary 6.2 (Conic transfer)**.** *If $t_2(2,q) \le q$, then*

$$\rho_{\mathcal{C}}(q) \le t_2(2,q). \tag{17}$$

More generally, every complete b-arc with $b \le q$ has a projective image disjoint from the prescribed conic.

Proof. Take $H = \mathcal{C}$, so $|H| = q+1$. For integral $b \le q$ one has $b(q+1) < q^2+q+1$, and Theorem 6.1 applies. □

Kim and Vu proved that, for all sufficiently large q, every projective plane of order q has a complete arc of size at most $\sqrt{q}(\log q)^c$ for an absolute constant c [12]. This is less than q for large q, so Corollary 6.2 gives the following immediate consequence.

Corollary 6.3. *For some absolute constant c and all sufficiently large prime powers q,*

$$\rho_{\mathcal{C}}(q) \le \sqrt{q}(\log q)^c.$$

The transfer does not exploit the conic and is unlikely to be sharp. It nevertheless supplies a uniform upper bound without having to preserve the arc condition during a separate random covering construction.

7 Even characteristic and the nucleus

Assume q is even and let ν be the nucleus of $\mathcal{C}$. The nucleus is outside $\mathcal{C}$, all tangents to $\mathcal{C}$ pass through ν, and every line through ν is one of these $q+1$ tangents.

Proposition 7.1 (The nucleus belongs to the arc)**.** *Let A be a k-arc disjoint from $\mathcal{C}$ with $\nu \in A$. Exactly $k-1$ A-secants are tangent to $\mathcal{C}$. In particular,*

$$I_{\mathcal{C}}(A) \ge k-1, \qquad I_{\mathcal{C}}(A) \equiv k-1 \pmod 2.$$

If A is $\mathcal{C}$-complete, then

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{k-1}{m}. \tag{18}$$

Proof. For every $a \in A \setminus \{\nu\}$, the line νa is tangent to $\mathcal{C}$, giving $k-1$ tangent secants. No line joining two points of $A \setminus \{\nu\}$ can be tangent: every tangent contains ν, which would place three points of A on one line. Hence these are exactly the tangent A-secants.

Every tangent contributes one to $I_{\mathcal{C}}(A)$, while every other line contributes zero or two. This proves the lower bound and parity statement. Inequality (18) follows from Corollary 5.1. □

Proposition 7.2 (The nucleus lies outside the arc)**.** *Let A be a $\mathcal{C}$-complete arc with $\nu \notin A$. Then*

$$1 \le r(\nu) \le m,$$

the tangent A-secants are exactly the secants through ν, and

$$I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2, \qquad I_{\mathcal{C}}(A) \ge r(\nu) \ge 1.$$

Consequently,

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}. \tag{19}$$

Proof. The nucleus is a required point outside $A \cup \mathcal{C}$, so relative completeness gives $r(\nu) \ge 1$; Lemma 2.1 gives the upper bound. A line is tangent to $\mathcal{C}$ exactly when it passes through ν. Thus the number of tangent A-secants is $r(\nu)$. Reducing the sum of the intersection sizes $|\ell \cap \mathcal{C}|$ modulo two gives $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2$. The remaining claims follow from Corollary 5.1. □

Corollary 7.3 (Universal even-characteristic loss)**.** *Every $\mathcal{C}$-complete arc in even characteristic satisfies*

$$I_{\mathcal{C}}(A) \ge 1.$$

Consequently, inequality (11) *can always be strengthened by replacing its final term by at least $-1/m$.*

Proof. If the nucleus belongs to A, Proposition 7.1 gives $I_{\mathcal{C}}(A) \ge k - 1 \ge 1$. Otherwise Proposition 7.2 gives $I_{\mathcal{C}}(A) \ge r(\nu) \ge 1$. □

These two cases are useful structural reductions, but Remark 5.3 limits what the incidence term alone can accomplish. For example, at $(q, k) = (16, 8)$ inequality (11) would be contradicted only by $I_{\mathcal{C}}(A) > 268$, whereas the universal upper bound is $I_{\mathcal{C}}(A) \le 56$. Thus the exact determination of $\rho_{\mathcal{C}}(16)$ below requires information beyond a lower bound on $I_{\mathcal{C}}(A)$ inserted into the present inequality.

8 Verified finite-field examples

The verified values at $q = 8, 9, 11$ use the lower bound $L_2(q)$ and explicit coordinates. At $q = 16$, the same ingredients give $8 \le \rho_{\mathcal{C}}(16) \le 9$; a checked projective classification excludes eight.

The algebraic rejection used in that classification is an instance of the following elementary linear-algebra observation. Linear systems of quadrics containing arcs are an established tool; see Glynn [10] and Ball [4]. Fix nonzero vector representatives of the projective points under consideration. If V is a finite-dimensional linear system of homogeneous forms, write $\mathrm{ev}_x \in V^*$ for evaluation at the representative of x. The zero or nonzero status of this evaluation is independent of the chosen representative.

Lemma 8.1 (Uncovered evaluation obstruction)**.** *Let A and U be finite sets of projective points. There is no nonzero $f \in V$ such that $U \subseteq Z(f)$ and $A \cap Z(f) = \varnothing$ if either of the following holds:*

1. *the evaluation map $f \mapsto (\mathrm{ev}_u(f))_{u \in U}$ is injective;*

2. *for some $a \in A$, the functional ev_a belongs to the span of $\{\mathrm{ev}_u : u \in U\}$.*

Proof. If f vanishes on U, injectivity in the first case gives $f = 0$. In the second case, applying a spanning relation for ev_a to f gives $f(a) = 0$, contrary to $A \cap Z(f) = \varnothing$. $\square$

For the full system of degree-d forms, the evaluation functionals are the degree-d Veronese images of the points, up to nonzero scalar. Thus the first alternative says that the uncovered Veronese images span the dual coefficient space; the second supplies a selected Veronese image already in their span. In particular, for the full degree-d system it is enough to exhibit $\binom{d+2}{2}$ uncovered points whose Veronese evaluation matrix is invertible. This formulation is not specific to conics.

The following sharp form is useful when a zero locus must avoid several selected points simultaneously. The vector-space covering threshold is classical; see, for example, Khare [14].

Proposition 8.2 (Finite-field evaluation dichotomy)**.** *Let W be a finite-dimensional vector space over $\mathbb{F}_q$, let $\nu : X \to W$ be any feature map, and let $U, A \subseteq X$ be finite with $|A| \leq q$. There is a linear form $f \in W^*$ such that*

$$f(\nu(u)) = 0 \quad (u \in U), \qquad f \neq 0, \qquad f(\nu(a)) \neq 0 \quad (a \in A)$$

if and only if

$$\langle \nu(U) \rangle \neq W \quad \textit{and} \quad \nu(a) \notin \langle \nu(U) \rangle \quad (a \in A).$$

The uniform threshold is sharp: $q + 1$ distinct hyperplanes through a common codimension-two subspace can cover W.

Proof. Put $S = \langle \nu(U) \rangle$. Necessity follows by applying f to the stated span relations. Conversely, $S \neq W$ makes the annihilator $S^\perp \subseteq W^*$ nonzero. For each $a \in A$, the condition $\nu(a) \notin S$ says that evaluation at $\nu(a)$ cuts out a proper hyperplane of $S^\perp$. At most q proper hyperplanes cannot cover a nonzero vector space over $\mathbb{F}_q$: their union has size at most $1 + q(|S^\perp|/q - 1) < |S^\perp|$. Any form outside their union has the required properties. For sharpness, quotient by a common codimension-two subspace and use all $q + 1$ lines in $\mathbb{F}_q^2$. $\square$

Taking ν to be the degree-d Veronese map gives the same exact dichotomy for homogeneous degree-d forms. Thus Lemma 8.1 is the one-selected-point exclusion mechanism, while Proposition 8.2 determines exactly when a single form can avoid as many as q selected points.

Proposition 8.3. *For the standard conic*

$$\mathcal{C} : \; XZ = Y^2$$

in the indicated projective planes,

$$\rho_{\mathcal{C}}(8) = 6, \qquad \rho_{\mathcal{C}}(9) = 6, \qquad \rho_{\mathcal{C}}(11) = 6.$$

Proof. Equation (12) gives

$$L_2(8) = L_2(9) = L_2(11) = 6.$$

Appendix A lists $\mathcal{C}$-complete arcs of size six in each plane. Their verification is described below. $\square$

Theorem 8.4 (Quadratic avoidance over $\mathbb{F}_{16}$)**.** *Let A be any eight-arc in* $\mathrm{PG}(2,16)$*, and let $U(A)$ be the points outside A that lie on no secant of A. There is no nonzero homogeneous quadratic form Q such that*

$$U(A) \subseteq Z(Q) \qquad \textit{and} \qquad A \cap Z(Q) = \varnothing.$$

The assertion includes singular quadratic forms.

Proof. Any four points of A form a projective frame. Normalize that frame and apply canonical augmentation. This gives respectively 4, 61, 454, and 2633 projective classes at sizes five through eight (from 182, 532, 5155, and 21495 legal extensions). Every retained transition is certified by an invertible 3×3 matrix together with pointwise projective scalar equalities, so the classification does not trust the canonical-labeling program. The final count of 2633 projective eight-arc classes independently reproduces Theorem 3.8.1 of Al-Seraji–Al-Ogali [2]; the count itself is not new. More specifically, each `StepBook.coverage` theorem says that every legal one-point extension of its parent is represented by a certified entry, and `StepBooksValid` says that the books cover the current parent list in exactly its listed order. Every entry carries a checked projective equivalence to a member of the next list. The theorem `classifiedAt_level8_of_frame` composes those equalities, and the frame-reduction theorem transports an arbitrary eight-arc into a listed leaf. Thus closure and exhaustiveness of the four class lists, not merely the arithmetic of each listed member, are kernel-checked; the C++ generator is not trusted for completeness.

For 2630 of the classified leaves, six points of $U(A)$ give an invertible 6×6 evaluation matrix, giving the first alternative of Lemma 8.1. In each of the remaining three leaves the evaluation rows have rank five, but an explicit linear combination places the evaluation functional of a point of A in their span, giving the second alternative.

Finally, the obstruction is projectively invariant. If gA is a normalized representative, replace Q by $Q \circ g^{-1}$. This is nonzero because g is invertible, its zero set is $gZ(Q)$, and projectivities carry secants to secants, so $U(gA) = gU(A)$. Thus a quadratic as in the statement would contradict the checked obstruction for the normalized leaf. The companion Lean development kernel-checks every local alternative; its global conic corollary additionally checks frame reduction and projective transport. □

Corollary 8.5 (Exact relative value over $\mathbb{F}_{16}$)**.** *For every nonsingular conic $\mathcal{C} \subseteq$* $\mathrm{PG}(2,16)$*,*

$$\rho_{\mathcal{C}}(16) = 9.$$

Proof. Equation (12) gives $L_2(16) = 8$. If a $\mathcal{C}$-complete eight-arc existed, its ordinary-uncovered locus would lie in the zero set of the nonzero quadratic defining $\mathcal{C}$, while the arc is disjoint from that zero set. This contradicts Theorem 8.4. The nine-point witness in Appendix A gives the matching upper bound. □

The published ordinary classification records arc counts, stabilizers, secant-index distributions, and conic-contained classes. We did not find the uncovered-locus rank split, Theorem 8.4, or its exact relative-conic corollary tabulated there.

Remark 8.6 (Anatomy of the three exceptional leaves). Using the binary $\mathbb{F}_{16}$ encoding from Appendix A and coefficient order $(X^2, Y^2, Z^2, XY, XZ, YZ)$, the one-dimensional kernels of the three exceptional evaluation matrices are generated by

$$(1,1,1,1,1,0), \qquad (0,0,0,5,4,1), \qquad (2,1,1,5,5,1).$$

The first and third forms split over $\mathbb{F}_{16}$ as

$$(X + 6Y + 6Z)(X + 7Y + 7Z) \quad \text{and} \quad 2(X + 4Y + 15Z)(X + 15Y + 4Z),$$

respectively, and each union of lines contains two selected points. The middle form is nonsingular and contains seven of the eight selected points. Thus the exceptional leaves are not merely unexplained rank defects: two are reducible quadratic configurations, while the third is an arc differing by one point from a seven-point subset of a conic. These descriptions follow by direct field arithmetic and are not used to establish classification completeness.

For reference, the verified incidence statistics are

q	$\|A\|$	$\binom{\|A\|}{2}$	$q^2 - \|A\|$	$I_{\mathcal{C}}(A)$	$\max_{x \notin A} r(x)$
8	6	15	58	16	3
9	6	15	75	13	3
11	6	15	115	0	3
16	9	36	247	32	4

The row at $q = 11$ has a stronger geometric interpretation. Since $I_{\mathcal{C}}(A) = 0$, none of the 15 secants meets $\mathcal{C}$: every secant of the witness is exterior to the conic. If N_i denotes the number of required points of index i, relative completeness and the maximum-index bound give only $i = 1, 2, 3$, while the two moment equations give

$$N_1 + N_2 + N_3 = 115, \qquad N_1 + 2N_2 + 3N_3 = 150, \qquad N_2 + 3N_3 = 45.$$

Consequently $(N_1, N_2, N_3) = (90, 15, 10)$.

Proposition 8.7 (The $q = 11$ code and extension spectrum)**.** *Let A be the six-point witness in Appendix A, and let C_A be its parity-check code. Then:*

(i) C_A is a projectively non-GRS $[6, 3, 4]_{11}$ MDS code of covering radius three, and its projective deep-hole syndrome locus is exactly the twelve-point conic $\mathcal{C}$;

(ii) the affine cosets of distances $0, 1, 2, 3$ number $(1, 60, 1150, 120)$, and the distance-two cosets split as $(900, 150, 100)$ according as they have one, two, or three minimum-weight leaders;

(iii) the simultaneous one-column MDS extensions form the independence complex of the icosahedral graph, with independence polynomial

$$1 + 12t + 36t^2 + 20t^3;$$

exactly six two-column and twenty three-column extensions are maximal, giving six complete eight-arcs and twenty complete nine-arcs containing the fixed seed, and there is no ten-arc extension; and

(iv) each witness column colours five pairwise disjoint edges of the icosahedron, missing an antipodal pair, and the six colour classes partition its thirty edges. Adding the six antipodal edges turns them into six perfect matchings of the augmented graph.

Proof. Every witness triple has nonzero determinant and the six columns span $\mathbb{F}_{11}^3$, giving the MDS parameters. The 6×6 evaluation matrix of the quadratic monomials

$$X^2,\ XY,\ XZ,\ Y^2,\ YZ,\ Z^2$$

is nonsingular, so no conic contains all six projective columns; by the preceding definition and the classical normal-rational-curve dictionary, the code is not projectively GRS. The formal theorem checks the no-conic premise; it does not build coding-theory equivalences into the trusted kernel statement. Direct finite arithmetic gives the secant-index spectrum

$$0 : 12, \qquad 1 : 90, \qquad 2 : 15, \qquad 3 : 10,$$

where index zero is precisely the conic. Multiplying each nonzero projective direction by its ten affine scalars gives all nonzero affine syndromes bijectively. In the companion formalization, membership in each counted set is equivalent to actual parity-check distance, and for every distance-two direction the determinant-zero witness pairs are proved equal to the supports of actual weight-two coefficient words. Uniqueness on an independent support and scalar multiplication then give the claimed coset and leader counts, not merely rescaled incidence counts. The index-zero directions give the covering radius.

For simultaneous extensions, two conic columns conflict exactly when their chord contains a witness column. No three conic columns are collinear, so there is no additional triple obstruction and valid extension sets are exactly graph-independent sets. Exhausting the 2^{12} subsets gives the displayed polynomial and maximal-set counts. The same determinant table partitions the thirty conflict edges into the six stated matchings. All these finite assertions are independently reproduced by the supplementary Python and C++ checkers and kernel-checked in the companion Lean module. Lean additionally proves that residual maximality upgrades to ordinary completeness: relative completeness confines every further point to the conic, and maximality excludes the remaining conic points. It also checks that the six missing antipodal edges are distinct and that adjoining the appropriate edge to each colour class gives a six-edge perfect matching; the six augmented classes are pairwise edge-disjoint and partition the augmented graph. □

The six-set is a classical Clebsch hexagon with icosahedral A_5 symmetry; this identification and the associated orbit geometry belong to prior art, not to the proposition's novelty posture [7, 9, 17]. More precisely, Dye's edge criterion at $q = 11$ and the complete-exterior-set formulation of Blokhuis–Seress–Wilbrink give $\mathcal{C}(\mathbb{F}_{11}) \subseteq U(A)$ [9, Theorem 6(iv), p. 281] [6, pp. 143, 146]. Dye's ten Brianchon concurrences and the chord-defect identity then give $|U(A)| = 12$, hence equality. This exact uncovered-locus conclusion is a short synthesis of classical inputs rather than a new classical configuration theorem. The exact conjunction of the relative-conic deep-hole locus, refined coset counts, and extension polynomial was not located in the bounded search, so we report it only as a checked synthesis rather than a priority claim.

Remark 8.8 (Auxiliary residual-game structure). There is also a game-theoretic consequence of this exceptional row, not used in any bound above. After the six witness points are occupied, all twelve points of $\mathcal{C}$ remain legal. Join two conic points when their chord contains a witness point. The resulting 12-vertex, 30-edge, 5-regular conflict graph is the icosahedral graph. Its antipodal involution is fixed-point-free, preserves adjacency, and pairs only nonadjacent vertices, so the usual copycat strategy makes the corresponding independent-set building position a P-position. The companion Lean development checks every seeded continuation, proves an exact recursive localization from the projective cap game to this independent-set game, and hence proves that the displayed six-point projective position itself is P. It also checks that the displayed $q = 9$ six-arc covers every projective point, including the conic, so that position is terminal P. These observations are consequences of the displayed witnesses, not novelty claims.

The supplementary Python program `verify_relative_conic_arcs.py` performs exact finite-field arithmetic and, for each witness:

(a) enumerates and normalizes all $q^2 + q + 1$ projective points;

(b) checks that $XZ = Y^2$ contains exactly $q+1$ points;

(c) verifies disjointness from the conic and that every pair-line contains exactly two selected points;

(d) verifies that every point outside $A \cup \mathcal{C}$ lies on a selected pair-line; and

(e) independently checks both equations in Proposition 2.2.

Its SHA-256 digest is

`e9508958d604e68c6c3d09fd3afadfaa8a3126508a51f1dfa993e7a7aed5d36a`.

The verifier is deliberately simpler than a search program: it certifies the listed upper-bound witnesses but makes no claim about search completeness.

Two further implementations independently regenerate Proposition 8.7. Their separately written source files are

`check_q11_structure.py`, `check_q11_structure.cpp`.

Their SHA-256 digests are

`0abe909c9aadce0db4c75f296c8de25e929dd1065c906996da8dec017e534d69`,
`1753674172d48f1d056d350e30baa9eb67de0810c84a96da0440947768ae041c`.

They agree on the projective, syndrome, chord, and group-action data. Both include transformed-coordinate invariance checks; the adversarial controls replace one witness point and mutate a group generator, respectively, and reject the exceptional conclusions. The small-field hyperplane checker `check_evaluation_dichotomy.py`, with digest

`6ed309bd2461ce9998cbd3bcaa5396379e6973b7503ce2dcdfb32c9806386566`,

exhausts every family of at most q projective hyperplanes for $q = 2, 3, 5$ and dimensions one through three, and checks the sharp $q+1$ plane cover.

The separate exact-classification generator `search_rhoc16.cpp` has SHA-256 digest

`589af8430e94b4c9c23ce895e6d32d2b3b5b9b387b1fb23ed6d3875cdee39031`.

Its frozen report `search_rhoc16_output.txt` has digest

`6989079b5cb64b0e57d5c42b872093fff99f861300b8fbb909daef450c15cc63`.

The generator emits local Lean certificates; its own assertions and canonical labels are not part of the trusted proof boundary.

The companion Lean library `lean/RelativeConicArcs/` formalizes the incidence, defect, conic, asymptotic, averaging, nucleus, evaluation-dichotomy, syndrome, and coding arguments. The downstream `Q11Coding` module kernel-checks the exact code parameters, covering radius, deep-hole locus, leader distributions, quadratic obstruction, chord partition, and extension complex. Its generic Boolean checker inspects conic disjointness, the arc condition, and coverage of the canonical projective representatives $[1 : y : z]$, $[0 : 1 : z]$, and $[0 : 0 : 1]$; a proved normalization and soundness theorem transports acceptance to relative completeness. The concrete checks use kernel reduction rather than `native_decide`. The trust manifest `lean/RelativeConicArcs/TRUST.md` records the verifier hash, the theorem map, and axiom profiles. In particular, the finite results do not depend on the external Kim–Vu input used for the conditional asymptotic upper bound.

9 Further questions

The elementary theory leaves several concrete problems.

Problem 9.1. Determine whether the uncovered-quadratic-rank obstruction used at $q = 16$ extends to other even orders or gives useful lower bounds on the size of a $\mathcal{C}$-complete arc beyond the scalar defect inequality.

Problem 9.2. Develop the higher-dimensional analogue for caps complete outside a prescribed quadric in $\mathrm{PG}(n, q)$. The evaluation-avoidance lemma already applies to arbitrary feature maps and every Veronese degree; the missing ingredients are the appropriate cap secant moments and useful examples in dimensions at least three.

Problem 9.3. Determine whether the exact defect remainder yields new lower bounds for arc-constrained saturating sets or the corresponding covering codes, and whether small-order scans beyond $q = 11$ produce further recognizable extension complexes. The evaluation formulation is also the elementary linear-algebraic interface used by polynomial-method arguments, but no slice-rank or cap-set consequence is asserted here.

Problem 9.4. Construct $\mathcal{C}$-complete arcs of size $O(\sqrt{q})$, or prove that this is impossible for an infinite family. Theorem 6.1 currently supplies only the general complete-arc bound $O(\sqrt{q}(\log q)^c)$.

Problem 9.5. Classify equality and bounded-defect configurations in Theorem 3.1. The equality pattern requires all nontrivial secant concurrence to occur at points of maximum possible index, a severe combinatorial restriction.

Problem 9.6. Determine whether the conic stabilizer or its coherent configuration yields useful finite-order restrictions not reducible to a bound on $I_{\mathcal{C}}(A)$. Remark 5.3 shows that an estimate for $I_{\mathcal{C}}(A)$ alone cannot improve the asymptotic constant obtained here.

The central distinction is between universal overlap, already measured by the classical second index equation, and genuinely conic-specific geometry. The former gives the lower bound in this paper. Progress beyond it will probably require an explicit construction, an algebraic obstruction, or a classification of near-extremal secant arrangements rather than another application of the same incidence budget.

A Witness coordinates and field encodings

All coordinates are homogeneous and use the standard conic $XZ = Y^2$.

For $q = 8$, write $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$ and encode $a_0 + a_1\alpha + a_2\alpha^2$ by the binary integer $a_0 + 2a_1 + 4a_2$. A witness is

$$(0, 1, 1), (0, 1, 2), (1, 0, 1),$$
$$(1, 0, 2), (1, 1, 2), (1, 1, 6).$$

For $q = 9$, write $\mathbb{F}_9 = \mathbb{F}_3[\alpha]/(\alpha^2 + 1)$ and encode $a_0 + a_1\alpha$ by $a_0 + 3a_1$. A witness is

$$(1, 0, 4), (1, 0, 5), (1, 1, 0),$$
$$(1, 1, 2), (1, 2, 3), (1, 2, 4).$$

For $q = 11$, coordinates lie in the prime field. A witness is

$$(1, 10, 0), (1, 9, 1), (1, 4, 7),$$
$$(1, 8, 5), (0, 1, 4), (1, 1, 7).$$

For $q = 16$, write $\mathbb{F}_{16} = \mathbb{F}_2[\alpha]/(\alpha^4 + \alpha + 1)$ and use binary coefficient encoding. A witness of size nine is

$$(1,5,14),(1,5,3),(1,9,2),(1,0,9),(1,13,13),$$
$$(1,6,10),(1,3,9),(0,1,11),(1,10,2).$$

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